2026 Tanner Crab CIE Review Management and Tier Considerations

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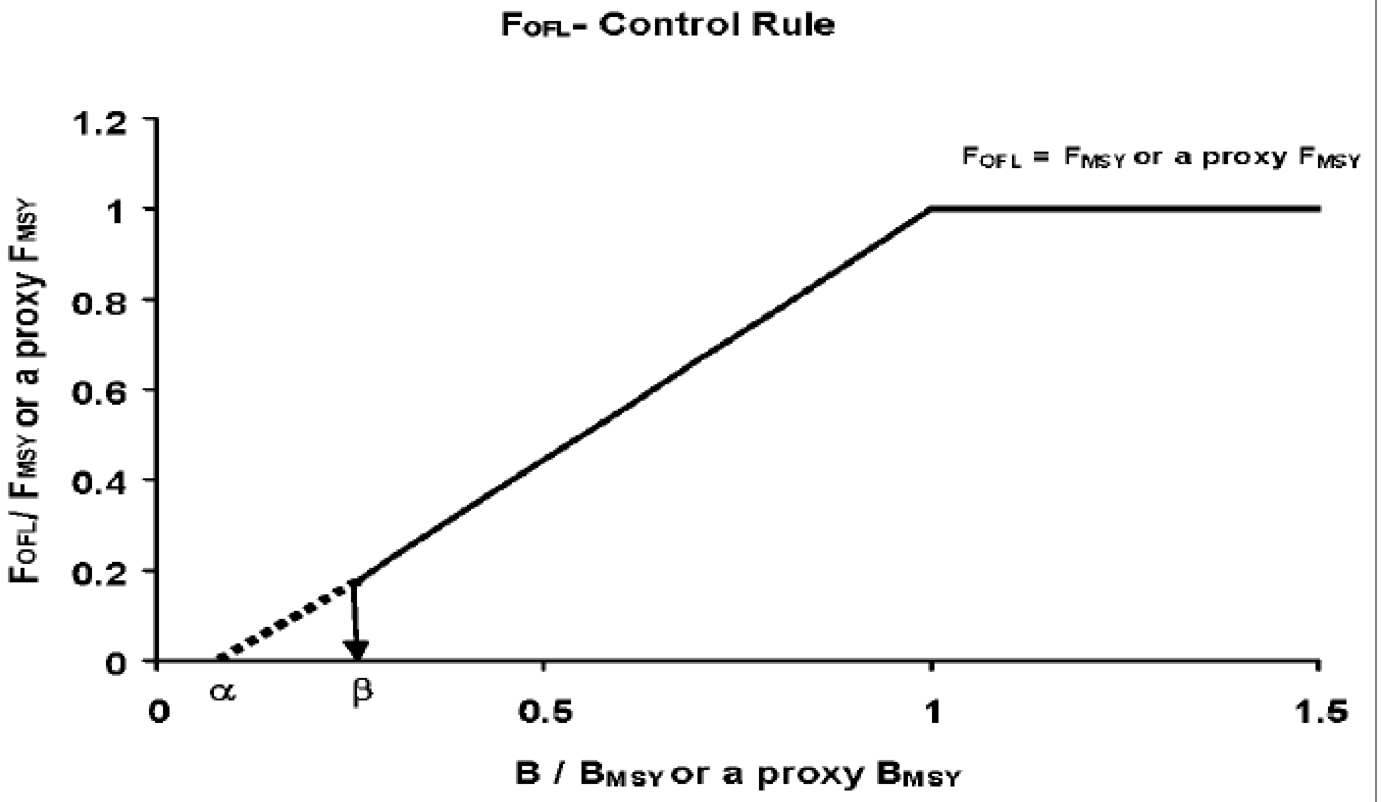
Crab Management Tiers 1-4

Tier 3

- no reliable S-R relationship but
- can estimate proxies for F_{MSY} and B_{MSY}
- proxies are SPR-related quantities
    - $B_{MSY}=B_{35\%}=0.35 \cdot B_{100}=0.35 \cdot \bar{R} \cdot SPR_{100}$
    - $F_{MSY}=F_{35\%}$
- originally developed for groundfish (Clark, 1983)
- the OFL accounts for all losses not attributable to *M*
- *maxABC* determined by p* = p(OFL>OFL_{TRUE}) = 0.49
- ABC typically determined by applying a buffer to OFL

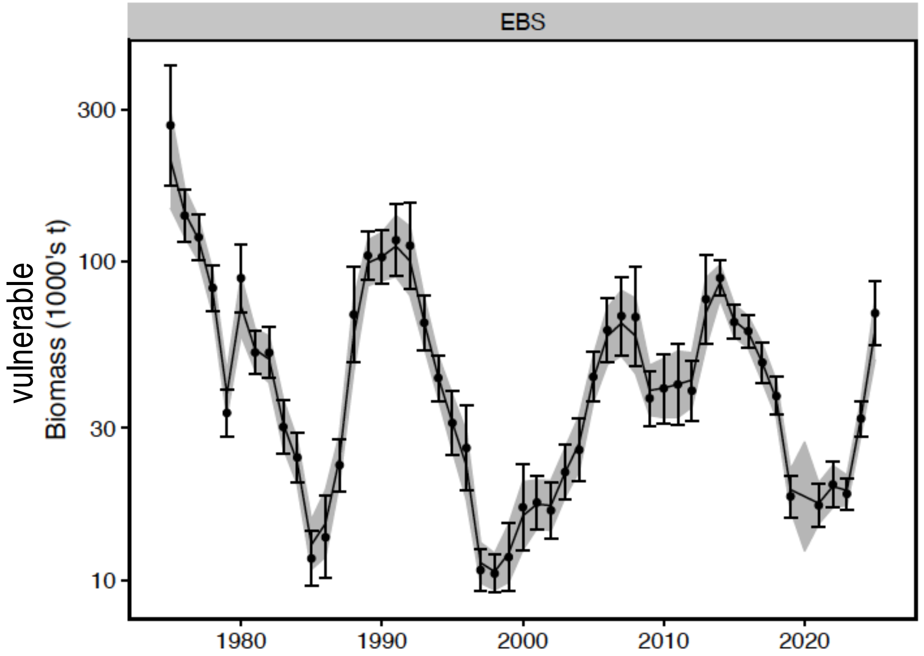


Tier 4

- life-history and recruitment info do not rise to Tier 3
- but reliable estimates of current survey biomass and *M* do
- proxies are:
    - $F_{MSY}=\gamma \cdot M$
    - $B_{MSY}=\bar{B}$

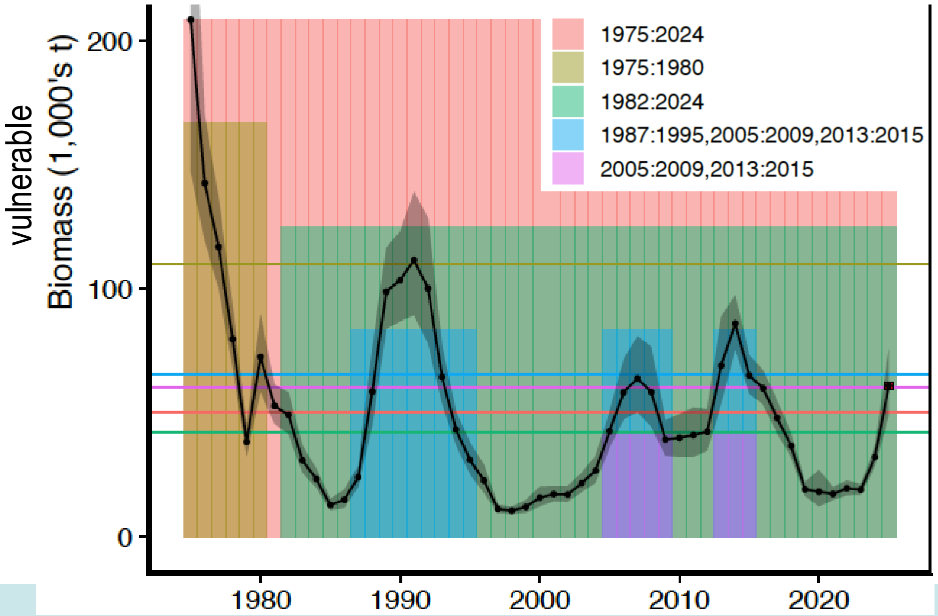
Tier 4 Backup Model

- M = 0.23 yr-*1*
- B and $\bar{B}$ based on rema-smoothed "vulnerable" biomass time series
- Potential averaging times for "vulnerable" biomass
  - 1982-2024
  - 1975-1980
  - 1975-2024
  - 1987-95, 2006-09, 2013-15
  - 2005-09, 2013-15



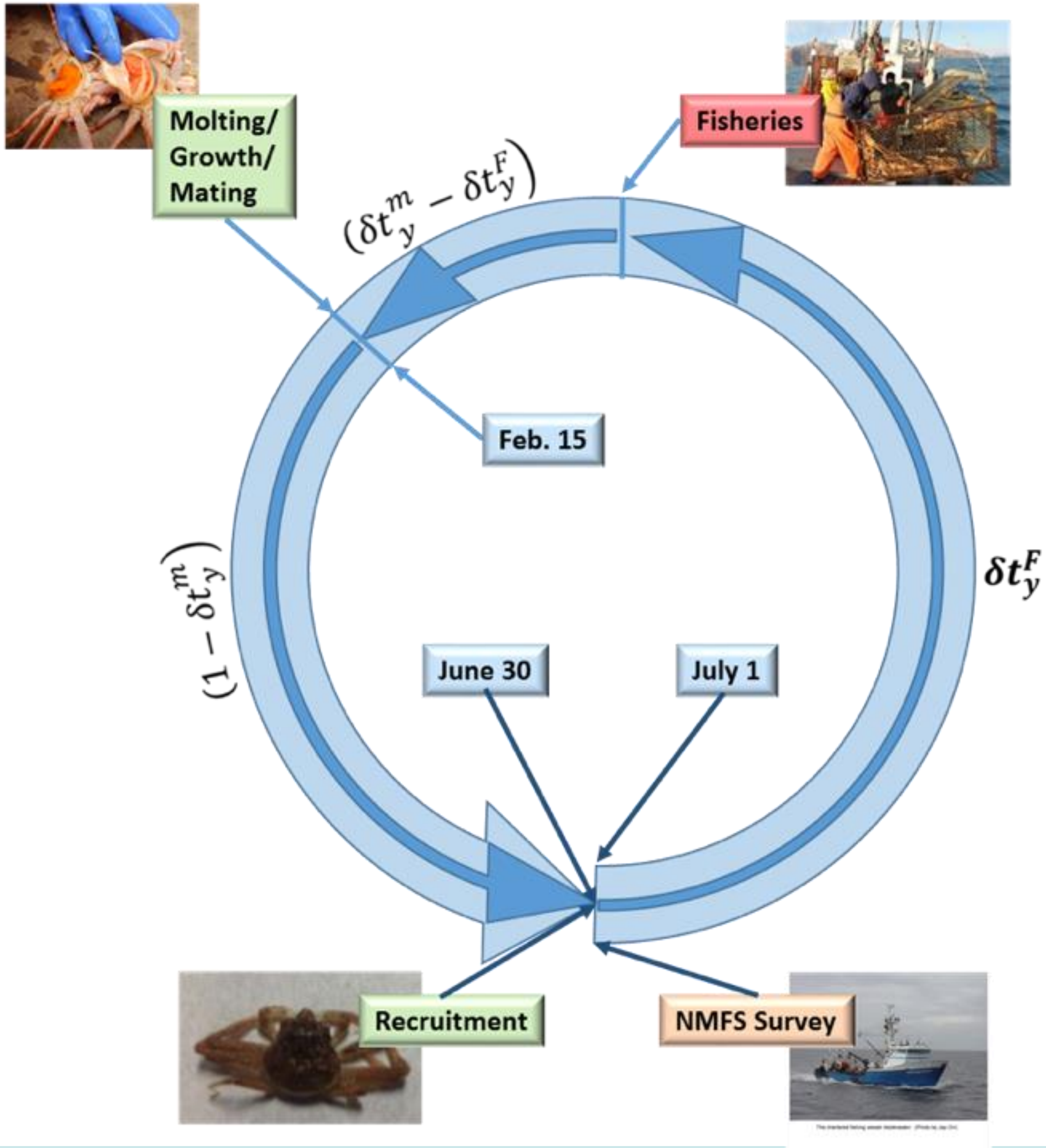
time block	M	B	B_{MSY}	status	F_{OFL}	OFL
1975:2025	0.23	61.04	50.50	1.21	0.23	12.54
1975:1980	0.23	61.04	109.90	0.56	0.12	6.71
1982:2025	0.23	61.04	42.34	1.44	0.23	12.54
1987:1995,2005:2009,2013:2015	0.23	61.04	65.88	0.93	0.21	11.62
2005:2009,2013:2015	0.23	61.04	60.42	1.01	0.23	12.54

biomass units: 1000's t

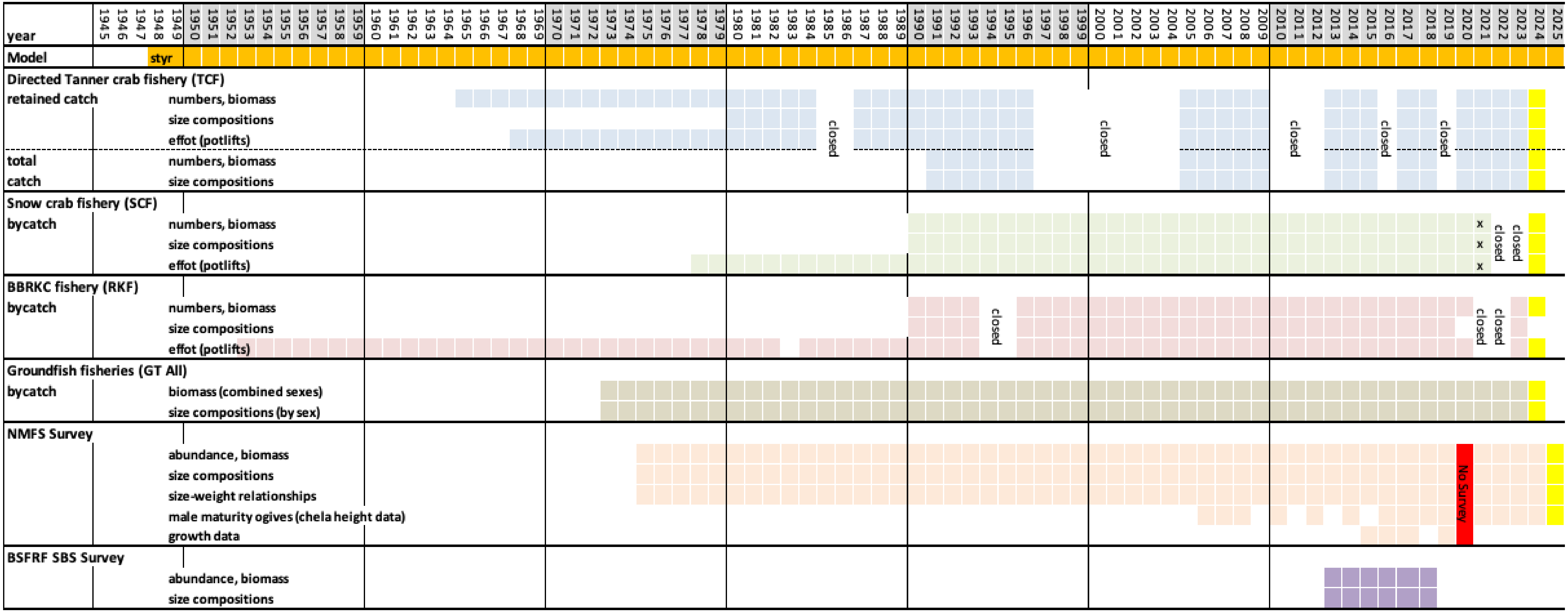


Tier 3 Assessment Model

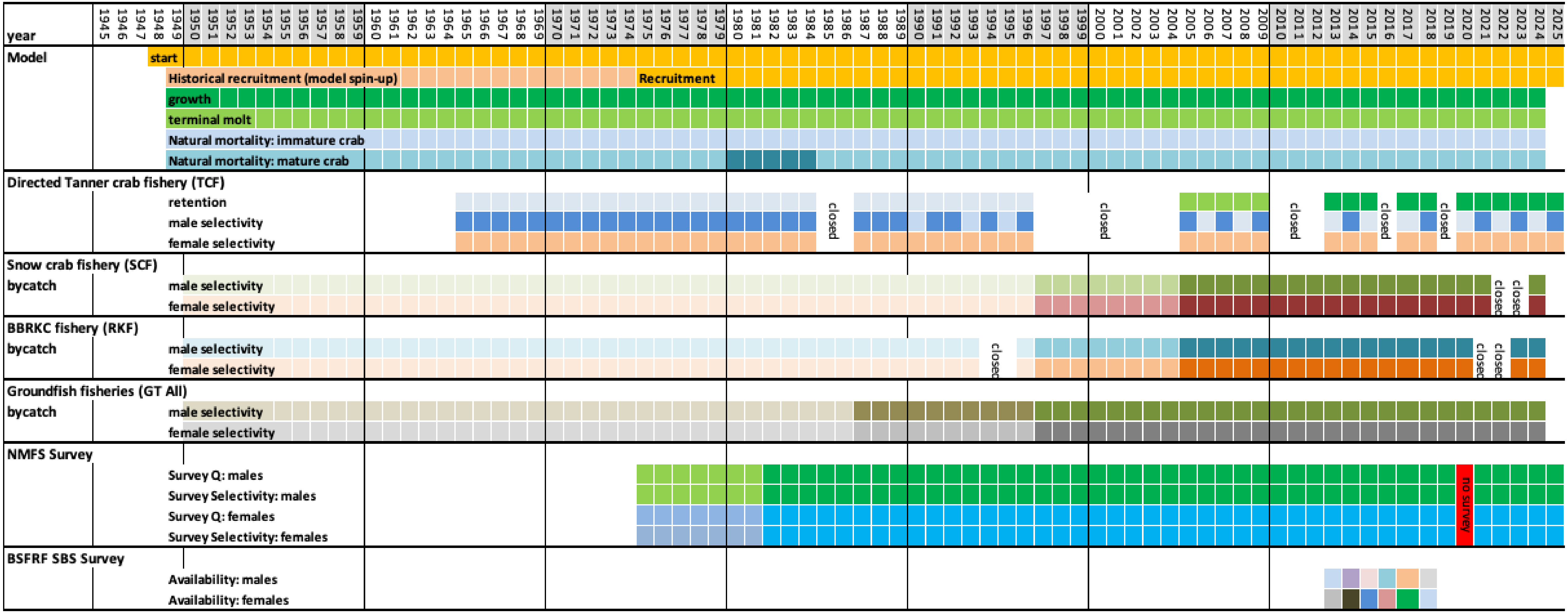
- Size-structured, two-sex model with terminal molt
  - Survey data
    - NMFS EBS shelf survey: 1975-present
    - BSFRF 2013-2018 side-by-side haul studies
    - growth data
    - male maturity data
  - Fishery data
    - directed fishery (areas combined)
      - retained catch
      - total catch
    - bycatch in
      - snow crab fishery
      - BBRKC fishery
      - groundfish fisheries
  - Estimates:
    - Annual recruitment
    - Annual numbers-at-size (M,F)
    - mature biomass (MMB, MFB)



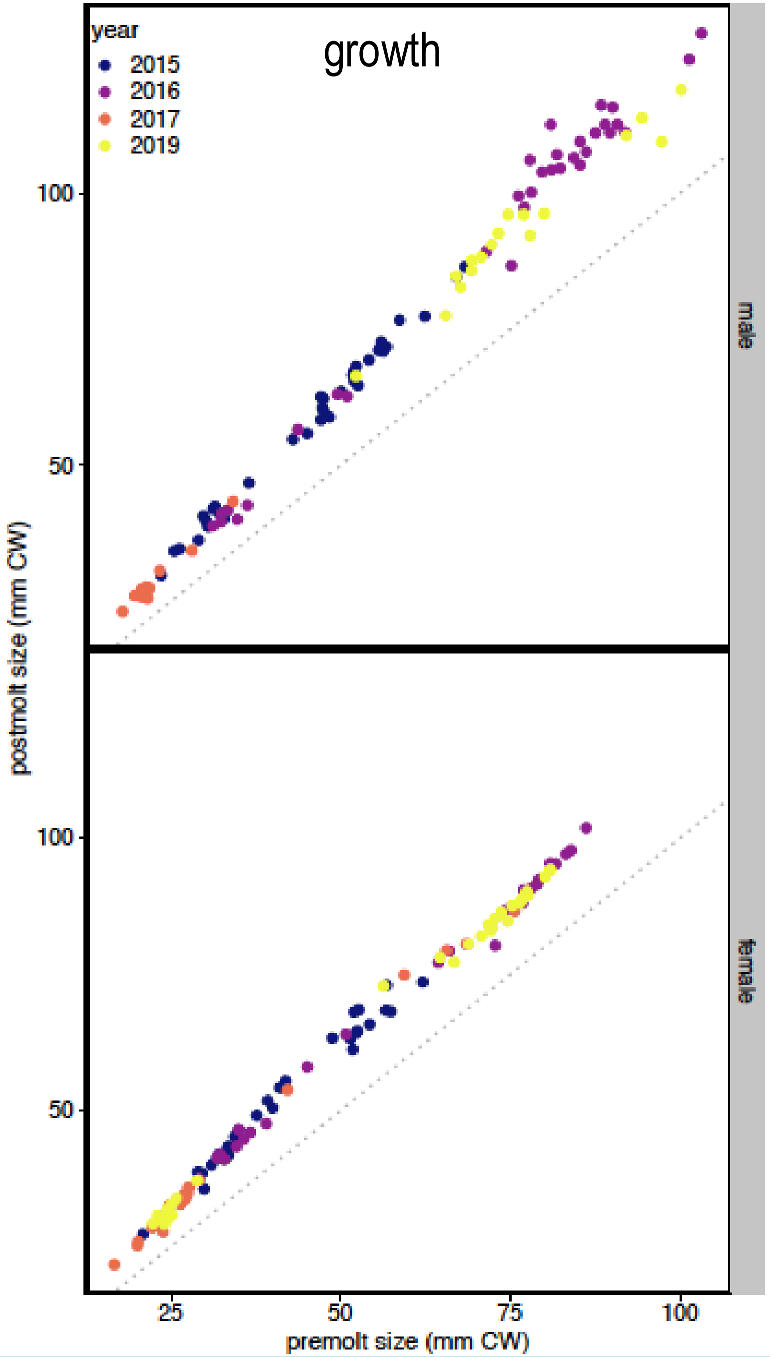
Assessment time frames: data

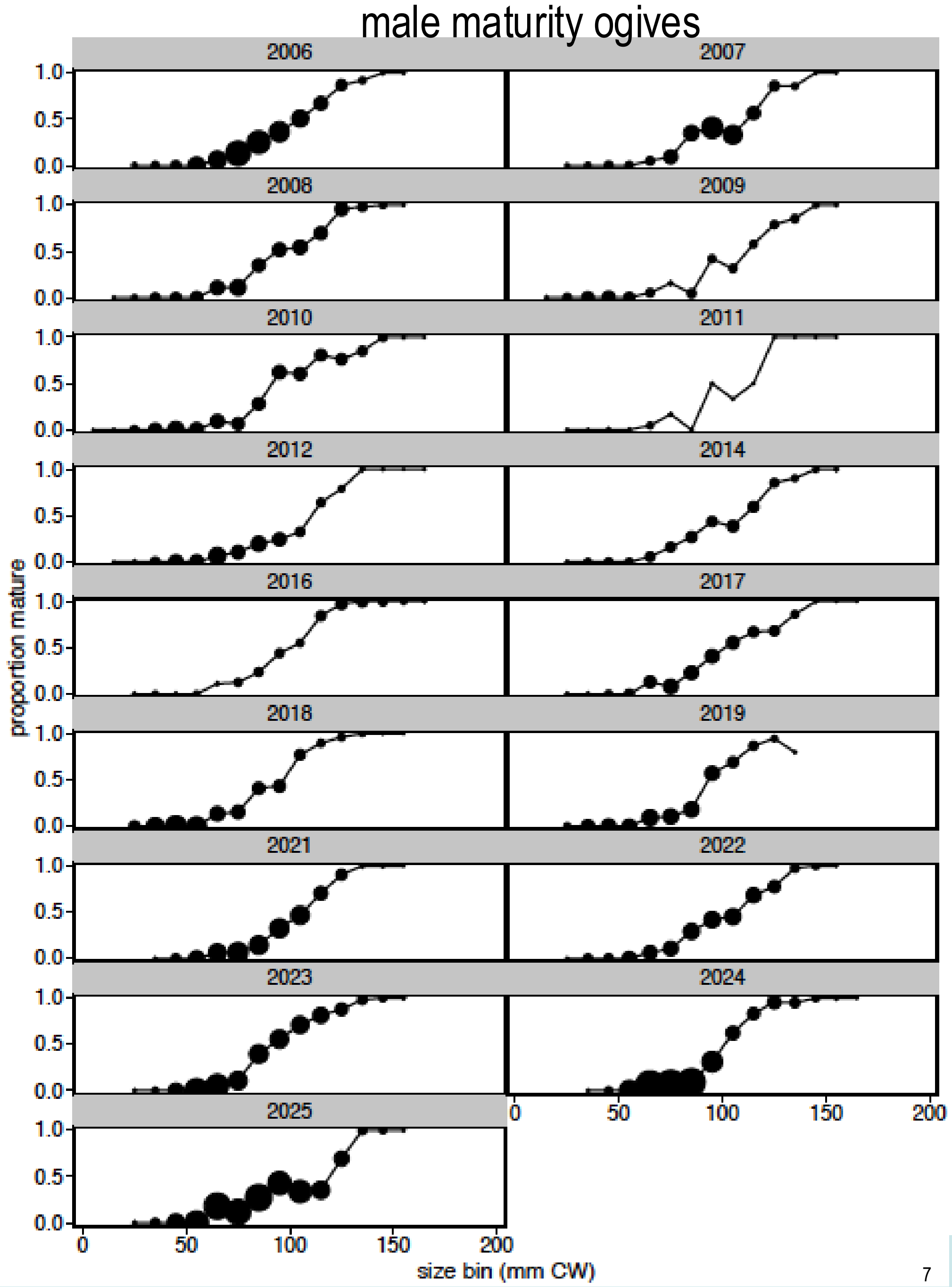


Assessment time frames: model processes



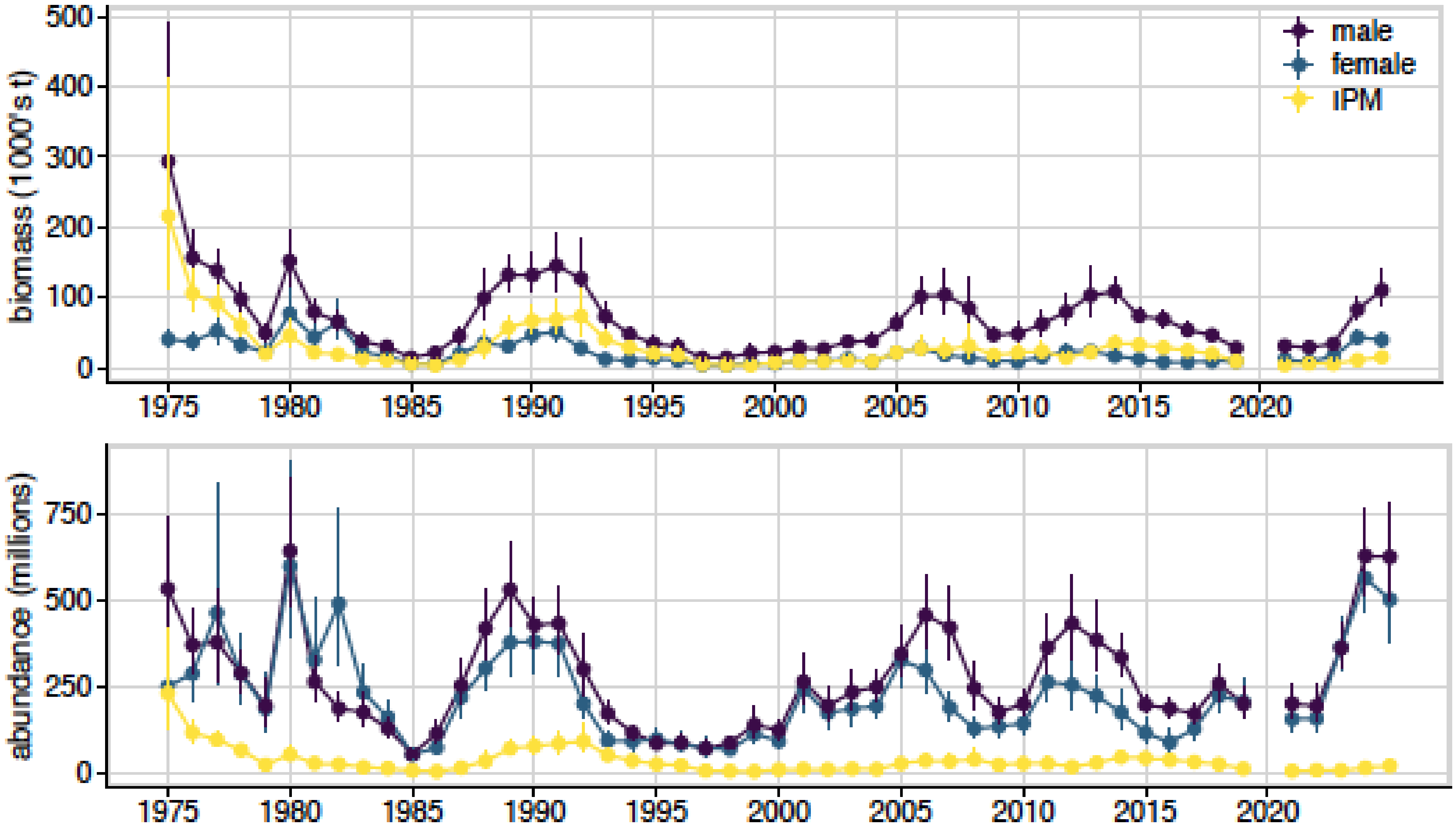
Life History Data





NMFS EBS Survey Data

IPM: Industry-preferred males (> 125 mm, 5 inch CW)

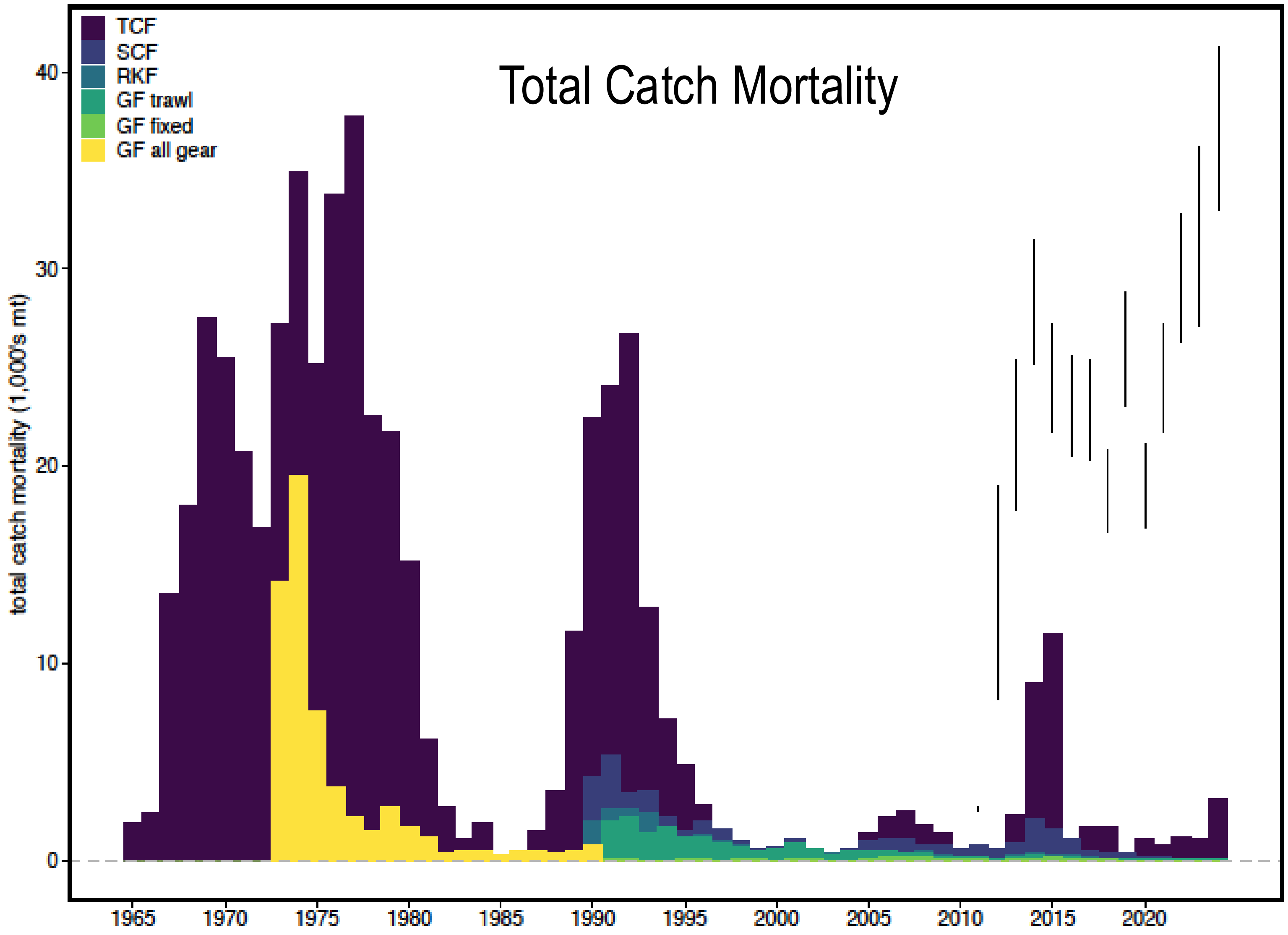


NOAA FISHERIES

OFLs and ABCs

vertical lines:

- upper end: federal OFL
- lower end: federal ABC



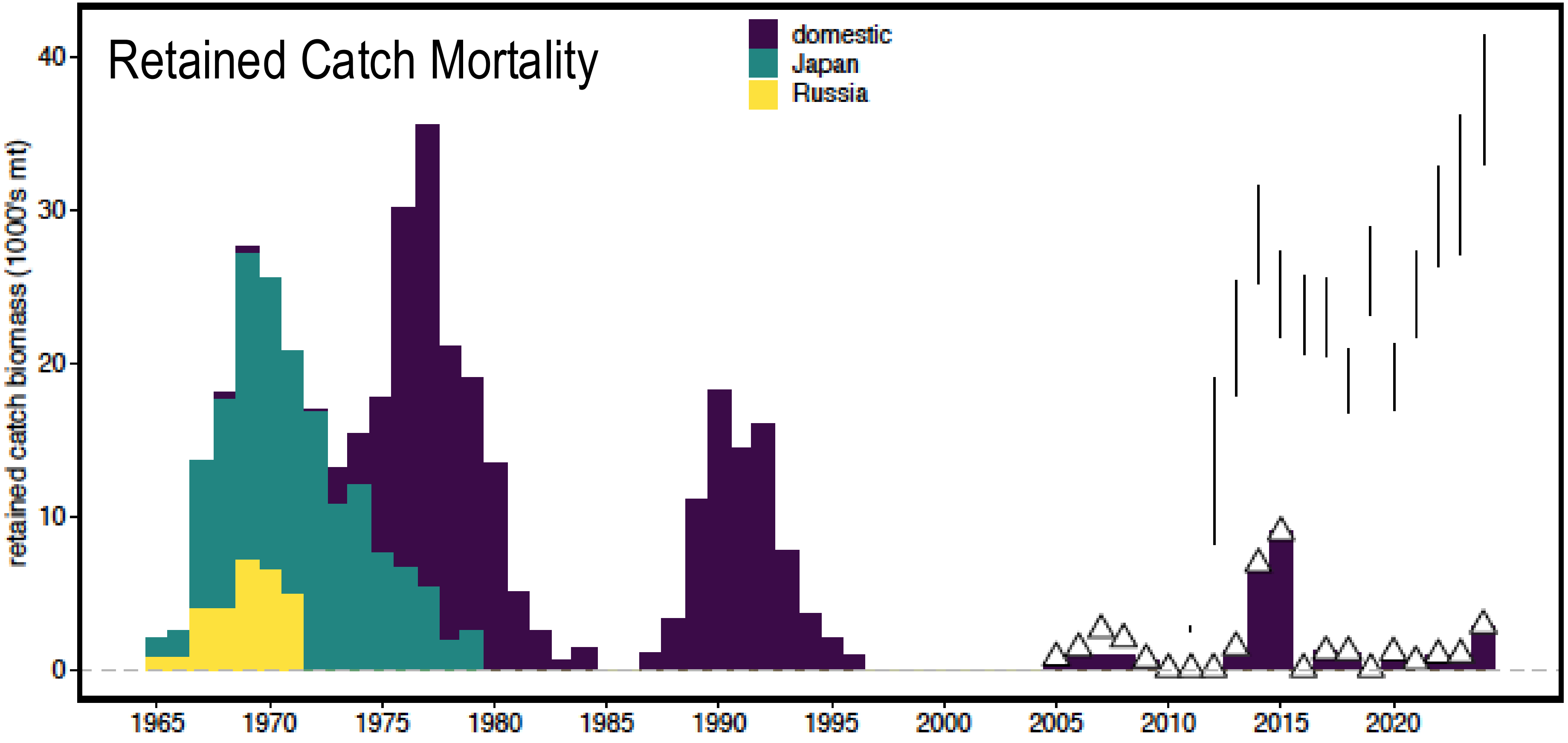
OFLs, ABCs, and TACs

vertical lines:

- upper end: federal OFL
- lower end: federal ABC

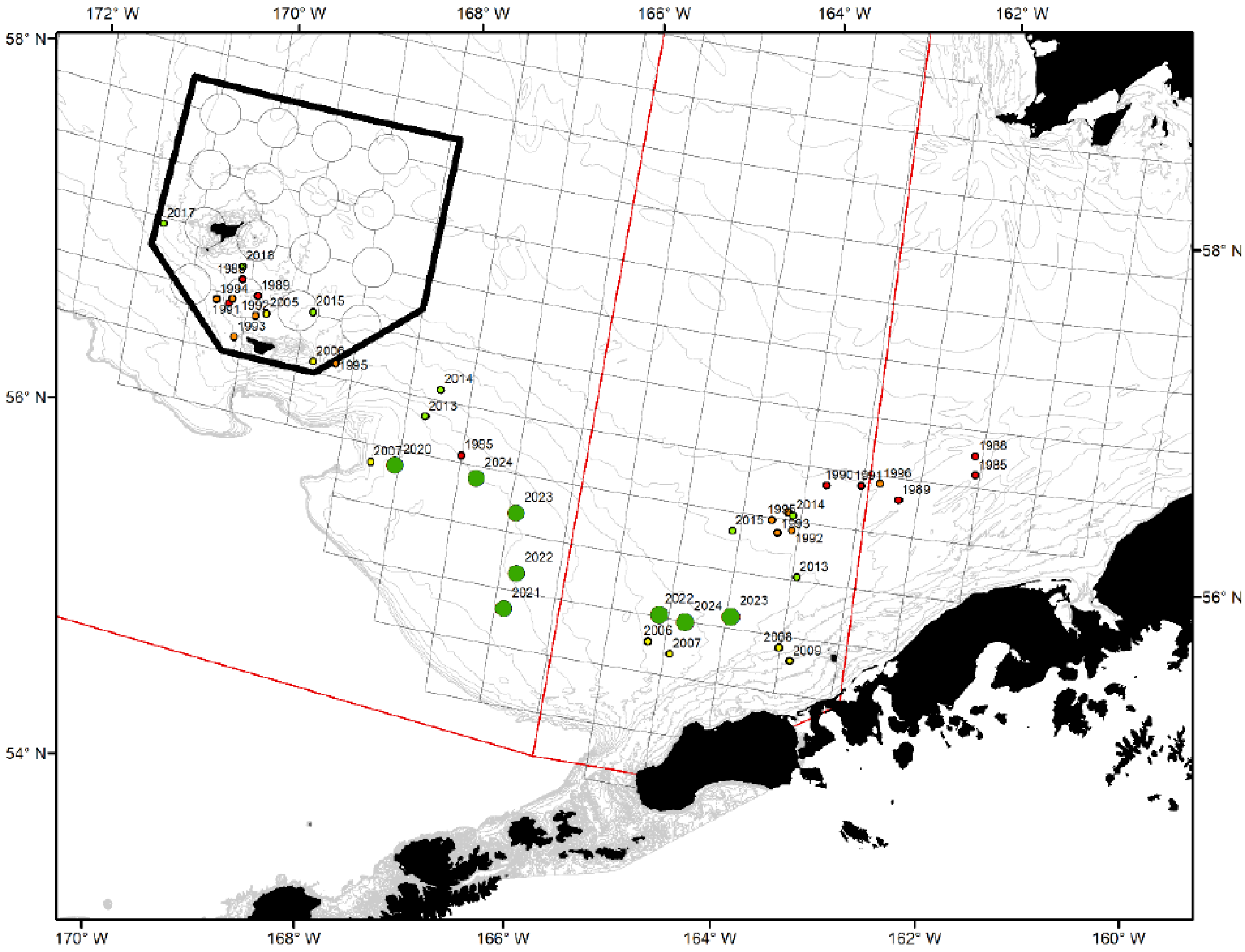
triangles:

- combined State TAC



Retained catch and effort

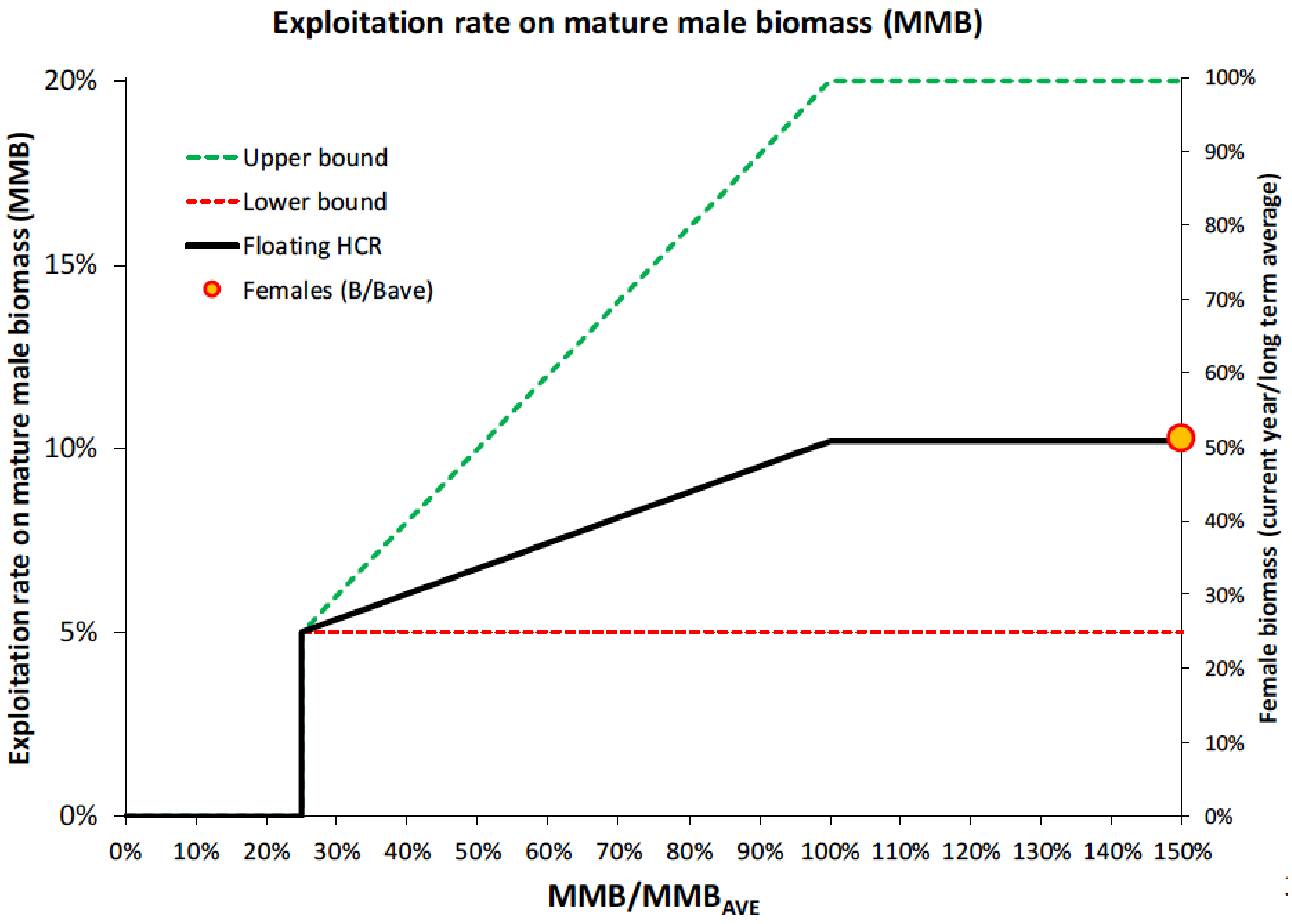
slide 8 from TAC setting



Courtesy of Ben Daly, ADFG

State Harvest Strategy

- Exploitation rates on MMB
  - 5% lower bound
  - 20% upper bound
  - Applied by State area (E/W 166W)
- "Floating" control rule determined by female biomass (entire EBS)
- Additional considerations: caps on exploitation rate for "exploitable" legal males
  - 100% new shell crab > 5in CW
  - 40% old shell crab > 5 in CW
  - cap: 50% by weight
- Requires:
  - MFB for entire EBS
  - MMB by E/W 166W
  - 5" male abundance by shell condition E/W 166W
  - Avg weight of 5" males E/W 166W



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